

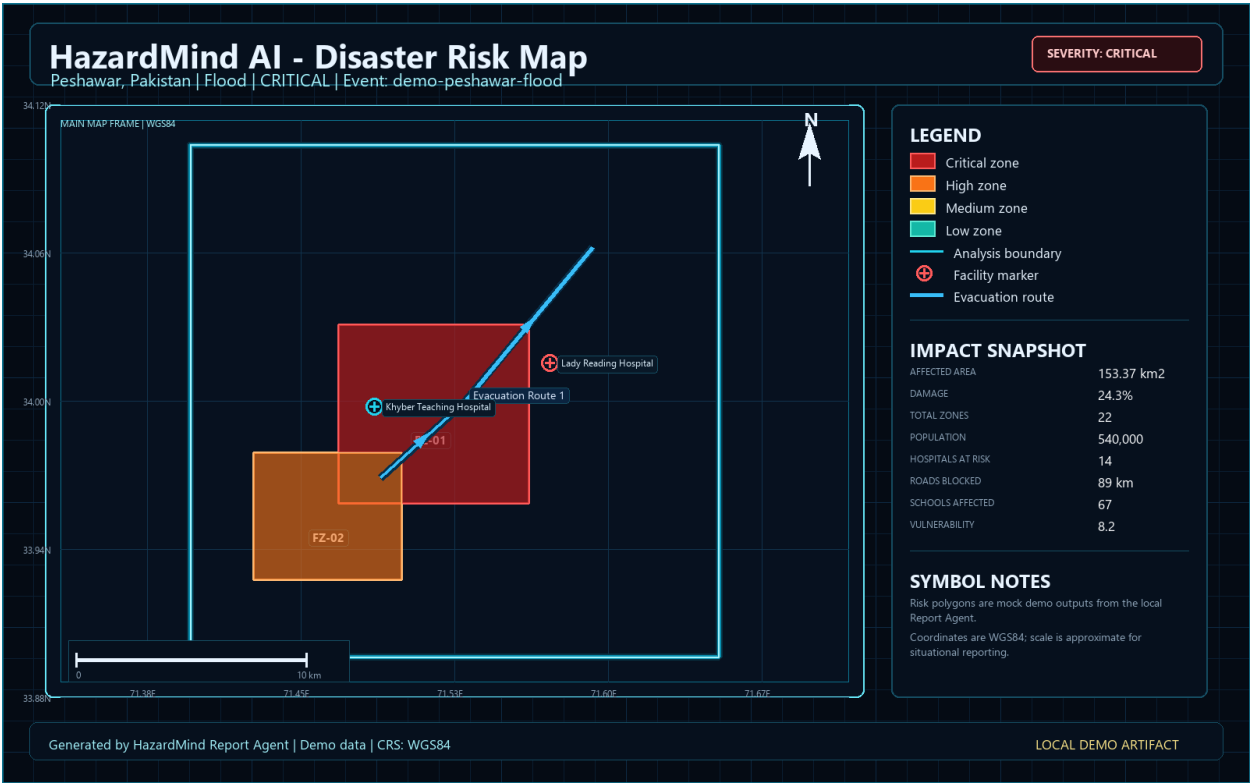
HazardMind AI Executive Disaster Report

Event ID	demo-peshawar-flood
Location	Peshawar, Pakistan
Hazard Type	Flood
Overall Severity	CRITICAL
Satellite Type	sentinel-1
Satellite Reason	cloud_cover_above_30_percent_sar_selected

Operational Statistics

Affected Area	153.37 km2
Damage Percent	24.3%
Total Zones	22
Population Affected	540,000
Hospitals at Risk	14
Roads Blocked	89 km
Schools Affected	67
Vulnerability Score	8.2

Generated Risk Map



Executive Summary

A CRITICAL flood has inundated 153.4 km2 of Peshawar, impacting 540,000 people and placing 14 hospitals, 67 schools, and 89 km of roads at risk. SAR analysis confirms the event at 91% confidence with deep-water concentrations in zones FZ-01 and FZ-02. The most urgent constraint is that only one evacuation route remains viable, creating a critical logistics bottleneck. Immediate priorities are evacuating high-risk zones, securing hospital access, and clearing blocked corridors. These are decision-support estimates requiring ground-truth validation; optical assessment is limited by 42% cloud cover.

Intelligence Assessment

Criticality	critical
Overall Confidence	91%
Escalation Required	True
Rationale	CRITICAL flood confirmed via SAR (91% confidence). 540,000 people affected, 14 hospitals at risk, 89 km
Key Decisions	Authorize immediate evacuation of high-risk flood zones FZ-01 and FZ-02; Establish emergency traffic con
Human Review Required	True

Map Narrative

The static risk map identifies a critical flood emergency in Peshawar, covering 153.37 km² (24.3% of the area). Using SAR imagery to penetrate cloud cover, the map highlights severe inundation across 22 zones, impacting 540,000 people and creating a critical logistics bottleneck with only one remaining evacuation route.

- 24.3% of the monitored area is inundated
- 89 km of road networks are blocked, severely restricting mobility
- High vulnerability score (8.2) driven by concentrations of affected schools and hospitals
- Single viable evacuation route creates a high-risk bottleneck for mass movement

Priority Timeline

Next 6 Hours	Evacuate high-risk flood zones FZ-01 and FZ-02; Secure access to Lady Reading Hospital; Establish emergency medical surge capacity
Next 24 Hours	Clear 89km of blocked roads for rescue access; Deploy medical surge teams to at-risk hospitals; Open temporary emergency shelters
Next 72 Hours	Conduct field validation of SAR damage estimates; Restore primary transportation corridors to affected schools and hospitals
Resources	Heavy machinery for road clearance; Water rescue boats and amphibious vehicles; Emergency medical supplies and personnel
Coordination	Manage evacuation bottleneck to prevent traffic gridlock; Sync satellite data with ground-truth reports; Coordinate with local authorities

Anomalies and Warnings

- MEDIUM: Report PDF and Map URLs are empty Handling: Verify document generation pipeline and update links
- LOW: Evacuation route count (1) is critically low relative to affected population (540,000) Handling: Flag as operational bottleneck for emergency logistics
- GeoJSON artifact URL is not available in local demo mode.
- Map URL is generated later by the local artifact step.
- Report PDF and Map URLs are empty; verify document generation pipeline.
- Satellite cloud cover at 42% limits optical ground-truth validation.

Quality Check

Status	not_ready
Event Id Present	True
Satellite Data Present	True
Hazard Data Present	True
Impact Data Present	True
Map Artifacts Present	True
Recommendations Present	True
Confidence Above Threshold	True
Warnings	GeoJSON artifact URL is not available in local demo mode.; Map URL is generated later by the local artifact step.
Blocking Issues	Critical data mismatch: evacuation_route_count is 0 in event payload but 1 in intelligence summary.; Missing ground-truth validation data.

Band-Ready Final Message

HazardMind Report Agent complete for demo-peshawar-flood (Peshawar, Pakistan, Flood). Criticality: critical; confidence: 91%. Summary: A CRITICAL flood has inundated 153.4 km² of Peshawar, impacting 540,000 people and placing 14 hospitals, 67 schools, and 89 km of roads at risk. SAR analysis confirms the event at 91% confidence with deep-water concentrations in zones FZ-01 and FZ-02. The most urgent constraint is that only one evacuation route remains viable, creating a critical logistics bottleneck. Immediate priorities are evacuating high-risk zones, securing hospital access, and clearing blocked corridors. These are decision-support estimates requiring ground-truth validation; optical assessment is limited by 42% cloud cover.

Detailed Incident Analysis

Satellite SAR analysis over Peshawar reveals a severe flood event with 153.37 km² of inundation affecting 24.3% of the monitored area. Deep water zones span 12.4 km² in Zone FZ-01, while Zone FZ-02 has 8.7 km² of high water. A population of 540,000 is impacted, with 67 schools affected, 89 km of roads blocked, and 14 hospitals at risk, including Lady Reading Hospital under high risk. Only one evacuation route remains viable, severely constraining response. The combination of extensive flooding, critical infrastructure at risk, and limited safe passage creates a CRITICAL emergency situation demanding immediate, large-scale intervention.

Technical Analysis

Sentinel-1 SAR was selected because cloud cover is 42%. The analysis uses SAR VV/VH ratio with mean value 0.24 and identifies 22 zones. Damage is estimated at 24.3%, with 89 km of roads blocked and a vulnerability score of 8.2.

Recommendations

- Prioritize evacuation in critical flood zones.
- Deploy emergency medical support near hospitals at risk.
- Clear blocked road corridors for rescue access.
- Open temporary shelters for displaced families.
- Monitor flood expansion using updated satellite imagery.
- Evacuate high-risk flood zones FZ-01 and FZ-02
- Secure access to Lady Reading Hospital
- Establish emergency traffic control on the single evacuation route
- Heavy machinery for road clearance
- Water rescue boats and amphibious vehicles

Response Priorities

- Activate evacuation support for FZ-01 and FZ-02.
- Protect hospital access and stage medical surge teams near high-risk facilities.
- Clear blocked road corridors needed for rescue and logistics movement.

Assumptions

- Satellite, hazard, and impact values are local demo data until upstream agents publish live outputs.
- Risk zone geometries are simplified mock polygons for dashboard and report demonstration.
- Facility and route coordinates are approximate and intended for prototype visualization.

Limitations

- No live Band messages are consumed in this local demo run.
- No field validation, hydrologic model, or live basemap tiles are used for artifact generation.
- Impact estimates should be treated as decision-support indicators, not official counts.

Model Source Note

Detailed report generated using Featherless Kimi K2.6. Executive summary generated using AI/ML API. Intelligence sources: criticality: featherless:deepseek-v4-pro, anomaly_check: featherless:gemma-4-31b-it, map_narrative: featherless:gemma-4-31b-it, priority_recommendations: featherless:gemma-4-31b-it, decision_brief: aiml:opus-4.8, quality_check: hybrid:deterministic+featherless:qwen-3.6-35b-a3b, band_ready_message: template+intelligence

Detailed Report	featherless:composite
Executive Summary	aiml:opus-4.8
Fallback Used	False
Featherless Model	moonshotai/Kimi-K2.6
Criticality	featherless:deepseek-v4-pro
Map Narrative	featherless:gemma-4-31b-it
Priority Timeline	featherless:gemma-4-31b-it
Quality Check	hybrid:deterministic+featherless:qwen-3.6-35b-a3b

Agent Log / Trace

Agent	Status	Timestamp	Message
hazardmind-report	complete	2026-06-13T18:03:00Z	Report Agent received event context
hazardmind-report	complete	2026-06-13T18:03:20Z	Featherless composite generated detailed disaster report
hazardmind-report	complete	2026-06-13T18:03:40Z	AI/ML generated executive summary
hazardmind-report	complete	2026-06-13T18:03:45Z	Criticality assessed
hazardmind-report	complete	2026-06-13T18:03:50Z	Anomalies checked
hazardmind-report	complete	2026-06-13T18:03:55Z	Map narrative generated
hazardmind-report	complete	2026-06-13T18:04:00Z	Priority timeline generated
hazardmind-report	complete	2026-06-13T18:04:05Z	Decision brief generated with Opus

Agent	Status	Timestamp	Message
hazardmind-report	complete	2026-06-13T18:04:10Z	Quality check completed
hazardmind-report	complete	2026-06-13T18:04:15Z	Band-ready final message prepared
hazardmind-report	complete	2026-06-13T18:04:00Z	Static cartography map generated locally